

Disproof of Conjecture 00000004340

earthking11

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Odd moments under central symmetry

Let μ be a centrally symmetric measure, so that it is invariant under the map $T(x) = -x$. Assume the relevant moments exist. For every $k \geq 0$, the change of variables $x \mapsto -x$ gives

$$\begin{aligned} m_{2k+1}(\mu) &= \int x^{2k+1} d\mu(x) \\ &= \int (-x)^{2k+1} d\mu(x) \\ &= -m_{2k+1}(\mu). \end{aligned}$$

Hence $m_{2k+1}(\mu) = 0$, and over the real numbers $m_{2k+1}(\mu) = 0$.

Proposition 1. *Two centrally symmetric measures cannot have different odd moments.*

Proof. Apply the calculation above to each measure. Every odd moment of each measure is zero, so their odd moments agree term by term. $\square$

The equality or inequality of the even moments is irrelevant to this obstruction. Moreover, a symmetry-breaking perturbation that makes an odd moment nonzero also makes the measure non-centrally-symmetric, so it cannot be the promised witness.

The Lean development formalizes the complete logical consequence for moment sequences and verifies cancellation on explicit finite paired supports.